

# PES University - Dept. of CSE

## Lab 1: Requirements Engineering & UML Use-Case Modelling

### Problem Statement #07 — Hostel Maintenance & Issue Ticketing System

SRN: PES1UG24CS612

| Req ID  | Type                                 | Description                                                                                                                            | Priority | Acceptance Criteria                                                                                                                                                                                     | Rationale (short)                                                       | Comments |
|---------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|----------|
| FR-001  | Functional                           | The system shall enable residents to log maintenance tickets with room number tags, photo attachments, and categorized issue severity. | High     | <b>Pass:</b> Ticket is created, assigned a unique ID, and routed to warden queue.<br><b>Fail:</b> Incomplete form accepted without room tag.                                                            | Core function for reporting hostel maintenance issues.                  |          |
| FR-002  | Functional                           | The system shall allow the Maintenance Warden to assign maintenance tickets to appropriate maintenance staff.                          | High     | <b>Pass:</b> A selected ticket is assigned to a maintenance staff member and the assignment is recorded.<br><b>Fail:</b> Ticket remains unassigned after a staff member is selected.                    | Ensures reported issues are dispatched to the responsible staff.        |          |
| FR-003  | Functional                           | The system shall allow maintenance staff to update the status of assigned tickets.                                                     | High     | <b>Pass:</b> Staff can change a ticket status to an available status such as In Progress or Resolved, and the new status is displayed.<br><b>Fail:</b> Status change is not recorded.                   | Allows residents and wardens to track maintenance progress.             |          |
| FR-004  | Functional                           | The system shall track the SLA timeline of each maintenance ticket.                                                                    | High     | <b>Pass:</b> The system records the ticket creation time and displays whether the ticket is within or beyond its SLA.<br><b>Fail:</b> SLA status is not updated according to the recorded time.         | Helps the administration monitor delayed maintenance work.              |          |
| FR-005  | Functional                           | The system shall allow residents to provide a closure sign-off for a resolved maintenance ticket.                                      | Medium   | <b>Pass:</b> A resident can confirm closure of a resolved ticket and the sign-off is recorded.<br><b>Fail:</b> The ticket is marked closed without recording the resident's sign-off.                   | Confirms that the reported issue has been satisfactorily addressed.     |          |
| NFR-001 | Nonfunc.<br>(Performance & Security) | The system shall trigger automated escalation SMS/Email alerts if high-priority tickets exceed a 24-hour SLA without status change.    | High     | Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.                                                                                                       | Ensures high-priority tickets are escalated when the SLA is exceeded.   |          |
| NFR-002 | Nonfunc.                             | The system shall protect maintenance ticket information so that only authorized users can access or modify ticket records.             | High     | <b>Pass:</b> Unauthorized users are denied access to ticket records, while authorized users can access permitted functions.<br><b>Fail:</b> An unauthorized user can view or modify ticket information. | Protects resident and maintenance information from unauthorized access. |          |